

# OpenDMC: An Open-Source Library and Performance Evaluation for Deep-learning-based Multi-frame Compression

Wei Gao\*  
gaowei262@pku.edu.cn  
Peking University Shenzhen Graduate  
School, China

Shangkun Sun  
sunshk@stu.pku.edu.cn  
Peking University Shenzhen Graduate  
School, China

Huiming Zheng  
hmzheng@stu.pku.edu.cn  
Peking University Shenzhen Graduate  
School, and Peng Cheng Laboratory,  
China

Yuyang Wu  
1979712607@qq.com  
Peking University Shenzhen Graduate  
School, China

Hua Ye  
yeh@pcl.ac.cn  
Peng Cheng Laboratory, China

Yongchi Zhang  
zhangych02@pcl.ac.cn  
Peng Cheng Laboratory, China

## ABSTRACT

Video streaming has become an essential component of our everyday routines. Nevertheless, video data imposes a significant strain on data usage, demanding substantial bandwidth and storage resources for effective transmission. To suit explosively increasing video transmission and storage requirements, deep-learning-based video compression has developed rapidly in the past few years. New methods have mushroomed in order to achieve better Rate-Distortion (RD) performance. However, the absence of an algorithm library that can effectively sort, classify, and conduct extensive benchmark testing on existing algorithms remains a challenge. In this paper, we present an open-source algorithm library called OpenDMC, which integrates a variety of end-to-end video compression methods in cross-platform environments. We provide comprehensive descriptions of the algorithms used in the library, including their contributions and implementation details. We perform a thorough benchmarking test to evaluate the performance of the algorithms. We meticulously compare and analyze each algorithm based on various metrics, including RD performance, running time, and GPU memory usage. The open-source library for OpenDMC is available at <https://openi.pcl.ac.cn/OpenDMC/>.

## CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; • **Software and its engineering** → **Open source model**; • **Theory of computation** → **Data compression**.

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## KEYWORDS

Video Compression, Open-source Software, Algorithm Library, Deep Learning

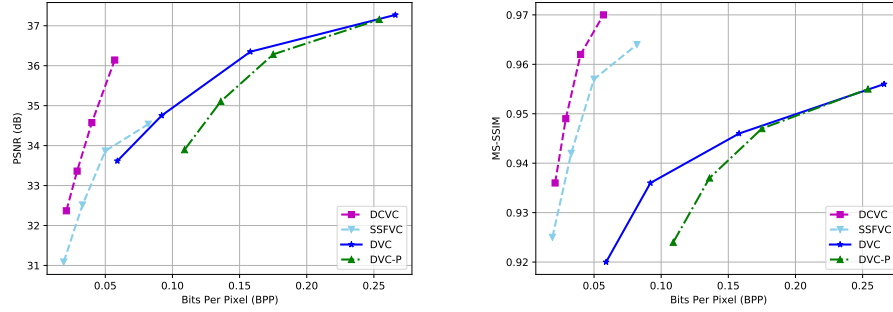
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## 1 INTRODUCTION

Video compression is a topic of momentous interest in the field of multimedia processing and has contributed significantly to the digital revolution in media and entertainment. The ultra high definition (UHD) video data leads to the great pressure on storage and transmission systems, and PP8K has been released as the latest 8K dataset [12]. The significance of video compression stems from its ability to reduce the storage and bandwidth requirements, while maintaining the high-quality video signal that is essential for the efficient transmission and playback of video content. Current mainstream compression techniques [1, 14, 15, 24] rely on exploiting the redundancy and statistical properties of video data. The video sequence is then analyzed using various spatial and temporal transforms, followed by quantization and entropy coding, and then compressed to a compact representation. Traditional video coding optimization research can be divided into two aspects, i.e., rate control [5–10] and low complexity [23] optimization, as well as the hardware implementations [11].

Over the years, video compression techniques have evolved from traditional codec-based approaches to deep-learning-based methods. Traditional codec-based approaches [17, 18, 20] often use a combination of different compression techniques, such as spatial and temporal resolution reduction, motion estimation, transformation, and quantization. With the advent of deep learning technologies, deep neural networks have emerged as a powerful tool for video compression. End-to-end approaches [1, 4, 13–15, 21, 22] have gained popularity in recent years. These methods use deep neural networks to learn and compress video data without requiring explicit pre-processing steps. One advantage of deep-learning-based compression methods is their ability to achieve high compression



**Figure 1: RD curve results. Rate-PSNR curve and Rate-MSSSIM curve tested in UVG dataset are plotted, respectively.**

ratios while maintaining acceptable levels of quality. In addition, these methods can reduce the computational complexity of video compression, making it possible to achieve real-time compression and decompression of video data. Moreover, the application of deep learning techniques to video compression has enabled novel video compression applications, such as compression of low-light and holographic videos.

Despite the emergence of an increasing number of deep-learning-based methods in video compression, there is still a significant challenge in developing an effective algorithm library capable of classifying, sorting, and rigorously benchmarking existing algorithms. Although there is a renowned library, CompressAI [2], which provides a framework for research on efficient neural networks for image compression, it has not provided sufficient algorithmic support in the field of video compression. Building an efficient algorithm library for video compression is still a relatively blank area. In this work, we propose the OpenDMC, an efficient, open-source algorithm library that supports diverse learning-based video compression methods across multiple platforms. OpenDMC evaluates and summarizes a variety of classic leading video compression algorithms while conducting extensive benchmark testing on multiple metrics, including RD performance, running time, and GPU memory usage.

In summary, our contributions are three-folded:

- To the best of our knowledge, we have developed OpenDMC, the first open-source deep learning algorithm library specifically designed for the video compression task, which bridges the lacuna that previously existed in this field.
- We have updated the pre-training process of a set of classical video compression algorithms across various platforms and organized these methods along different dimensions.
- We have incorporated a diverse range of evaluation metrics into OpenDMC including RD performance, runtime, and GPU memory usage.

## 2 SUPPORTED ALGORITHM LIBRARY

**DVC.** DVC [15] has pioneered the use of deep-learning-based video compression algorithms, providing a classical reference framework for emerging subsequent works. Different from traditional

compression methods, DVC implements all key components of video compression, such as motion estimation, motion compensation, residual compression, motion compensation, quantization, and bitrate estimation, via end-to-end neural networks. DVC applies an encoder-decoder optical flow convolution network to estimate inter-frame motion, and then compresses the residual to save bit rate. The reference frame is warped to the target frame via the predicted flow and then the residual is derived. Then the residual is passed into non-linear neural networks for better quantization. To make the quantization differentiable, the quantization operation is approximated as a process of adding uniform noise in the training stage. In Entropy coding, a CNN is used to better estimate the probability distributions of generated latent representations. In supervision, the optimization objective is to minimize the number of bits for encoding and reduce the distortion between the original frame and the re-constructed frame. The loss function is the weighted sum of MSE loss (for measuring distortion) and the bit count used for encoding the representations. Benefiting the jointly end-to-end optimization of modules of video compression, DVC attains compelling performance compared with traditional methods.

**SSFVC.** SSFVC [1] aims to enhance the motion compensation module, which faces drawbacks in traditional 2D flow-based compensation approaches including: 1) Divergence in optimization objective. 2) Divergence in supervision strategy. 3) Limitation on disocclusions. 4) The additional complexity and constraints to designing compression networks. To address these, SSFVC proposes the scale-space flow estimation and scale-space warping techniques. A scale field is added as a third dimension to the conventional 2-channel flow field to allow for better handling of difficult cases and a more graceful degradation when flow-based predictions are not possible. The scale dimension enables the model to learn to adaptively blur the source content before warping based on how accurately it predicts the subsequent frame. This approach results in a smaller intermediate residual error and a more compressible residual, since the model does not need to devote as many bits to correcting errors introduced by the warping process.

**DCVC.** Different from previous compression methods that use residual encoding, DCVC [14] utilizes learnable high-dimensional temporal contextual features as conditions for frame compression.

**Table 1: BD-PSNR and BD-MSSSIM comparisons with DVC [15] in different test sequences for each algorithm.**

| Sequence \ Algorithm |               | DVC-P   | SSFVC  | DCVC   |
|----------------------|---------------|---------|--------|--------|
| Metric               |               |         |        |        |
| BD-PSNR (dB)         | Beauty        | -0.19   | 0.55   | 1.18   |
|                      | ReadySteadyGo | -0.74   | -2.23  | 3.91   |
|                      | ShakeNDry     | -0.38   | 1.76   | 2.51   |
|                      | YachtRide     | 1.42    | 1.24   | 5.85   |
|                      | HoneyBee      | -0.27   | -0.70  | -1.99  |
|                      | Bosphorus     | -0.29   | 0.94   | 0.97   |
|                      | Jockey        | -1.20   | 0.61   | 0.45   |
|                      | Average       | -0.23   | 0.19   | 1.78   |
| BD-MSSSIM            | Beauty        | -0.0023 | 0.0489 | 0.0602 |
|                      | ReadySteadyGo | -0.0041 | 0.0161 | 0.0570 |
|                      | ShakeNDry     | -0.0053 | 0.0658 | 0.0739 |
|                      | YachtRide     | -0.0078 | 0.0589 | 0.0908 |
|                      | HoneyBee      | -0.0002 | 0.0278 | 0.0120 |
|                      | Bosphorus     | -0.0022 | 0.0322 | 0.0331 |
|                      | Jockey        | -0.0031 | 0.0222 | 0.0299 |
|                      | Average       | -0.0035 | 0.0367 | 0.0492 |

To derive contextual information, DCVC first transforms previously decoded frames from the pixel domain to the feature domain using a feature extractor, while using motion estimation modules to learn motion vectors. In this case, high-dimensional features can provide richer temporal information than traditional pixel-domain predicted frames, and different channels offer greater flexibility in extracting different types of information to achieve better reconstruction of high-frequency details in the current frame. The derived motion vector guides feature extraction after encoding and decoding. To address the spatial discontinuity caused by motion compensation, a context refinement module is applied to generate the final contextual features. These contextual features are then used as conditional inputs to both the encoder and decoder within a parallel and concatenated architecture. In contrast to residual coding, which is highly inefficient in areas with large motion or newly appearing content, where inter-frame correlation is low and residual energy is high, DCVC’s conditional coding adaptively explores spatial correlation to encode these areas, and obtains flexible transformations for both intra-frame and inter-frame coding.

**DVC-P.** DVC-P [24] aims to improve video compression performance from the perspective of quality evaluation. Motivated by the observation that optimizes video compression solely for improving PSNR does not always enhance the perceptual quality of decoded videos, DVC-P propose a deep video compression framework with perceptual optimizations. In specific, inspired by generative adversarial networks, DVC-P optimizes the DVC framework [15] with a discriminator network and mixed loss. The mixed loss is a weighted sum of adversarial loss, MSE loss, VGG-based loss, and bit rate loss. Besides, DVC-P further eliminates checkerboard artifacts in DVC through nearest-neighbor interpolation. Checkerboard artifacts result from the de-convolutions in DVC that can cause colorful blurs in the decoded video. Thanks to the joint optimization, DVC-P attains significant improvements in both the BD-rate and visual experience.

### 3 BENCHMARKING RESULTS AND ANALYSIS

#### 3.1 Datasets

To conduct large-scale testing for OpenDMC, we choose UVG dataset [16] as the test dataset. The UVG dataset encompasses seven high frame rate videos, each with a resolution of 1920×1080 pixels. The adjacent frames in these videos exhibit strong correlation facilitating inter-frame prediction for better testing performance.

#### 3.2 Metrics for Performance Evaluation

We adopt PSNR and MS-SSIM [19] as the evaluation metrics in RD performance benchmark test. PSNR represents the signal fidelity of compressed video while MS-SSIM measures the structural similarity between two videos, reflecting the subjective visual quality of human eyes. In order to evaluate the time complexity of each algorithm, we employ running time during the inference period. Moreover, GPU memory occupancy is used to represent the space complexity of each algorithm.

#### 3.3 Rate-Distortion Performance Evaluation

For qualitative comparison, the RD performance of each algorithm in the library is shown in Fig. 1. DCVC obtains the best performance in both metrics, benefiting from the efficient conditional coding framework. The second best is SSFVC, which indicates that the optical flow network brings great gain to the compression framework. DVC and DVC-P have similar RD performances.

For a quantitative comparison, we calculate BD-PSNR and BD-MSSSIM [3] comparison with DVC for other algorithms in the library. Test results in the seven sequences are presented independently. It is shown that each algorithm performs quite differently on different test sequences. Each algorithm performs better on *YachtRide*. In some special cases, such as *HoneyBee*, DCVC and SSFVC perform worse than expected, which indicates that the adjacent frames in *HoneyBee* cannot be fully referenced. Relatively, DCVC, SSFVC and DVC-P obtain greater performance gain on BD-MSSSIM metric than BD-PSNR metric.

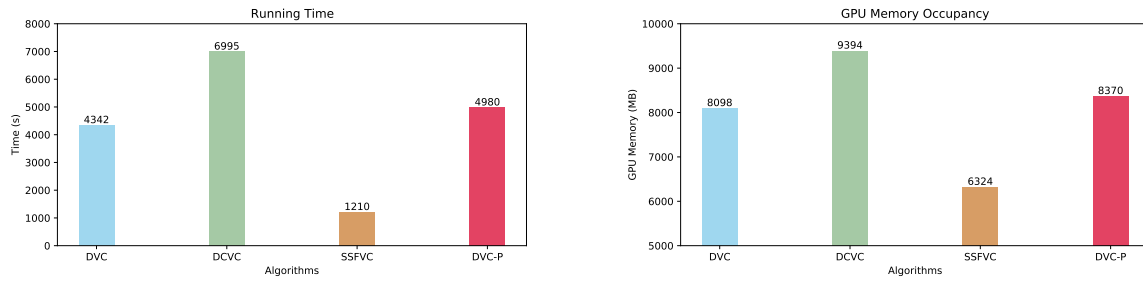


Figure 2: Comparison of Running Time and GPU memory occupancy results.

### 3.4 Running Time and GPU Memory Occupancy Evaluation

The running time results are presented in Fig. 2. Noted that the running time is the total result of the frames in all the sequences of UVG dataset. It is obvious that the fastest algorithm is SSFVC due to its scale-space warp. The auto-regressive model predicting the context adopted in DCVC significantly increases the time complexity, which makes DCVC get the longest running time.

GPU memory occupancy is shown in Fig. 2. SSFVC achieves the least GPU memory occupancy, benefiting from the lightweight architecture. DCVC possesses the largest GPU memory usage, due to its deep network architecture and complicated entropy model. Due to the similar architecture of DVC-P and DVC, the running time and GPU memory occupancy are similar.

## 4 CONCLUSION

In this paper, we release the first deep-learning-based video open-source algorithm library for video compression. In addition, the algorithm library is built in cross-platform environments. First, the algorithm used in the library is succinctly introduced. Paradigm design and module function introduction are demonstrated in this paper. Afterwards, we conduct performance test for the representative deep-learning-based video compression algorithms. We elaborately analyze the performance of each model in terms of RD performance, time complexity and space complexity. We sincerely hope the open source algorithm library can accelerate the development of deep-learning-based video compression.

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